

Project Green-PVC	Client Mundra Petrochem Ltd.	Plant Code MPL-OEPMC	Document Identification Number OEPMC-HVA-0000-EC-00001	Contract No. 66-6753
	General Engineering Specification for HVAC			
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7.20.3 Hot Insulation

7.20.3.1 Insulation for personnel protection is applicable where exposed surface temperatures exceed 65°C in normal and short term operating conditions. Insulation is not desired for piping and equipment for which heat loss is desired except for personnel protection and to avoid thermal stress problem. Bare surfaces of pipe/equipment shall be provided with one coat of 20 micron min. (dry film thickness) heat resistant primer up to 120°C operating temperature.

7.20.3.2 Resin bonded mineral wool conforming to IS-8183 shall be used for hot insulation with thermal conductivity not to exceed 0.43 mW/cm °C at 50 °C mean temperature. Preformed pipe sections/mattresses of min. density 140 kg/m3 /128 kg/m3 respectively shall be used. The minimum thickness of insulation shall be as per Appendix-1.

Comply

7.20.3.3 Moisture barrier shall be provided for application up to 125 °C. A 3 mm wet thick coating of mastic/ShaliKote shall be applied to the surface of the insulation as soon as possible after erection, to reduce the time the insulation is exposed to the weather to a minimum. Whilst this coat is still wet, glass cloth shall be laid over the surface and embedded in the mastic/ShaliKote. Care shall be taken to ensure that the glass cloth is laid smooth and free from wrinkles and that no pockets of air are trapped beneath the surface. At junctions in the glass cloth, the overlap shall not be less than 75 mm. A second 3 mm thick coat of mastic/ShaliKote shall be applied after approx. 12 hours. When dry, this coating shall be a minimum of 1.5 mm thick. Care must be taken, however, to ensure that the individual coats are not greater than 3 mm (especially in comers), otherwise some cracking of dried coat may result. The total dry film thickness of moisture seal shall not be less than 2.5 mm with a tolerance of -0% to +50%. The mastic/ShaliKote shall not be applied over wet insulation or until the adhesive is dry. During this drying time, the insulation shall be protected from the weather by tarpaulin or similar materials.

Noted

7.20.3.4 Hot insulation shall be cladded with 26 SWG aluminium sheeting.

7.20.4 Cold Insulation

7.20.4.1 The pipes shall be insulated and finished as per Appendix-2 and the minimum thickness of insulation shall be as per Appendix I.

7.20.4.2 The ducts shall be insulated and finished as per Appendix-3.

7.20.4.3 Exterior insulation of duct shall consist of fibre-glass duct wrap with thermal conductivity not to exceed 0.3 cal. cm/hr. cm2 °C (0.24 Btu in/ft2 hr. °F or 0.035 W/m-K) at 23.8°C mean temperature differential. Duct wrap to have, factory applied, fire rated aluminium foil and vapour barrier. Insulation shall be wrapped tightly on the duct work with all circumferential joints butted with longitudinal joints overlapped a minimum of 50mm.

Comply

7.20.4.4 Pipe insulation material shall be polyurethane foam with thermal conductivity not to exceed 0.019 KCal/hr. m°C (0.16 Btu in/ft2 hr. °F) at 10°C mean temperature after ageing.

Comply

7.20.4.5 Vapour barrier shall be provided for application up to 125° C. A 3 mm wet thick coating of vapour seal mastic/ShaliKote shall be applied to the surface of the insulation as soon as possible after erection, to reduce the time the insulation is exposed to the weather to a minimum. Whilst this coat is still wet, glass cloth shall be laid over the surface and embedded in the mastic/ShaliKote. Care shall be taken to ensure that the glass cloth is laid smooth and free from wrinkles and that no pockets of air are trapped beneath the surface. At junctions in the glass cloth, the overlap shall not



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Noted

be less than 75 mm. A second 3 mm thick coat of mastic/ShaliKote shall be applied after approx. 12 hours. When dry, this coating shall be a minimum of 1.5 mm thick. Care must be taken, however, to ensure that the individual coats are not greater than 3 mm (especially in corners), otherwise some cracking of dried coat may result. The total dry film thickness of moisture seal shall not be less than 2.5 mm with a tolerance of -0% to +50%. The mastic/ShaliKote shall not be applied over wet insulation or until the adhesive is dry. During this drying time, the insulation shall be protected from the weather by tarpaulin or similar materials.

- 7.20.4.6 The application, insulation material and recommended thickness are given in Appendix-1.
- 7.20.5 Acoustic Insulation
 - 7.20.5.1 Duct acoustic insulation shall be provided on the inner surface of ducting for a minimum distance of 6 m from AHU / Package Air-conditioning Unit outlet.
 - 7.20.5.2 Duct liner material shall have a flame spread rating not to exceed 25 and a smoke developed rating not higher than 50 (NFPA 255 or ASTM E 84). Duct lining should not flame, glow, smolder, or smoke when tested in accordance with ASTM C 411 at a temperature of 121°C.
 - 7.20.5.3 Insulation shall not absorb more than 1.13% moisture by weight at the end of 72 hours exposure to air at 49°C and 96% relative humidity. Thermal conductivity shall not exceed 0.3 cal cm/hr cm² °C (0.24 Btu in/ft² hr. °F or 0.035 W/m-K) at 23.8°C mean temperature differential. Material shall not have a surface coating and shall not erode at velocities of 20 metres/sec. (4000 fpm).
 - 7.20.5.4 The liner shall be adhered to the duct sheets before assembly using a rubber based adhesive spread over the entire surface. Ducts over 600mm shall have liner additionally secured with welded nails and speed clips on 300mm centers in both directions. It shall be covered by 0.5 mm perforated aluminium sheets having 3 mm perforation at 6 mm centres.
 - 7.20.5.5 For all inspection covers and hatches on equipment, pump casing, valve bodies & flanges more than 100 mm dia, the insulation shall be applied so as to facilitate easy removal and replacement maintaining continuity of vapour seal.

Comply

7.21 Controls And Instruments

- 7.21.1 Vendor shall include all controls & instruments as required for safety & operational requirements of the plant. Also, Vendor to prepare Cause & Effect diagram for activations of various valves/dampeners, start/stop of HVAC system with respect to gas detection & Fire alarm system. The contractor shall prepare common cause & effect diagram of overall F&G system.
- 7.21.2 Repeat interface signal like room temperature, humidity, pressure and corrosion monitoring of each respective control building (LCR, PIB, MCR & CCR) from HVAC system shall be wired to Owner supplied DCS system for monitoring. Owner supplied F&G system shall provide only potential free contact output to HVAC PLC system to control any components of HVAC system in case of any Fire or Gas high alarm. As per SIL verification report suitable voting logic shall be considered in F&G system for shutdown/trip signal to HVAC system

Fire & Gas detector of HVAC fresh air duct shall be wired to Owner supplied F&G system for monitoring and controlling of inlet damper.
- 7.21.3 Main HVAC Control Panel
 - 7.21.3.1 Automatic operation of all controls, wherever feasible, is required, with manual overdrive for

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Classification: **Confidential**

B 66-6753-11064918 EC 000 05 IA 26.04.2023